Kairos Lab Report

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Classifying a post's performance relative to its author's usual level: `below`, `typical`, `above`. The primary metric is macro-F1, which weights all three classes equally. The reference level is the median of `log(1 + reactions)` over the author's earlier posts.

Methods Used in the Final Model

- **Tabular features:** post length, type and formatting, links, hashtags, mentions, emojis, letter case, punctuation; time and its cyclical encoding, post age, author profile and followers. These capture presentation, timing and audience context.
- **Author history:** reaction statistics, windows of 3/5/10/20 posts, trends, exponential averaging, deviations from usual formatting and posting rhythm. These capture typical behaviour and recent changes. Additional history features use only strictly earlier posts, including initial unlabelled posts.
- **Text:** normalization of anonymized identifiers; TF-IDF of words/bigrams and character 3-5-grams. Word features use post text plus attachment titles/transcripts and reshare text; character features capture fragments within post text. Normalization retains placeholder types without individual codes.
- **Embeddings:** frozen multilingual MiniLM (384 features); splitting long texts into chunks, averaging and L2 normalization. These vectors capture semantic similarity beyond word overlap. Encoder weights remain unchanged during training.
- **History retrieval:** BM25, cosine similarity of embeddings, statistics of similar past posts and alignment with the author's profile. BM25 finds lexical matches; embeddings find semantic matches. Past reactions indicate how similar content performed for that author.
- **Images:** RGB histograms for colour distribution and frozen CLIP ViT-B/32 (512 features) for learned visual features; vectors are averaged per post. SVM uses images/previews from core; RGB and regression CLIP use original post images. Missing images receive zero vectors and an availability indicator.
- **Preprocessing:** median imputation for missing numerical values, StandardScaler for the relevant model inputs, One-Hot Encoding and LightGBM categorical features. Learned preprocessing is fitted on the corresponding training fold.
- **Base models:** LinearSVC combines sparse text, numerical and embedding features; HistGradientBoostingRegressor predicts the deviation of `log(1 + reactions)` from the author's level; LightGBM models nonlinear interactions; cosine kNN (`k=15`, class-wise sums of similarities) uses nearby training-post embeddings.
- **Stacking:** logistic regression on OOF (out-of-fold) predictions from **3 author-based folds**. Each training post receives predictions from models trained without its author's fold. At inference, fold outputs are averaged before the meta-model. Class weights and adjustments to log probabilities balance class decisions.
- **Reproducibility:** disjoint author-based splits, fixed seeds and library/model versions, embedding caching. `python my_model.py` trains the fixed configuration and writes predictions and metrics. Missing pretrained weights are downloaded at pinned revisions; caches are rebuilt as needed.

Rejected and Replaced Variants

Numerical hyperparameter searches are grouped. Checks included pilots and full CV; not all excluded methods were initially unsuccessful. Selection prioritized overall macro-F1, code simplicity and correspondence to course methods, rather than improvement in one class alone.

Method / Approach	What Was Tested and Why It Was Not Retained
SVD / LSA	TF-IDF compression into a smaller latent feature space; replaced by SVM, and subsequent checks did not bring the method back into the ensemble.
RBF / Nyström	Nonlinear transformation using a kernel approximation instead of a purely linear representation; did not improve the metrics.
Lemmatization, Unicode/HTML normalization	Reducing words to base forms and additional text cleanup; no CV gain over the simpler preprocessing.
χ^2 selection, quantile discretization	Selecting terms associated with labels and encoding numerical values into frequency-based intervals; not selected.
LDA	Adding topic probabilities as document features: a small CV gain, but worse `evaluate` results.
VADER, pronouns, NLI emotion/intent features	Lexicon-based sentiment/style signals and semantic emotion or intent features; no sufficient gain.
Additional text context	Lexical similarity to history was reverted after `evaluate`; joint encoding of current and historical text and subtracting the author's mean vector did not justify the complexity.
SIFT / Bag of Visual Words	Local visual descriptors: averaging them worsened CV; a visual-word vocabulary initially helped but was later removed during simplification because it did not provide a substantial gain.
Weighted late fusion	Combining separate model outputs with selected weights: text-based LR with boosting was replaced by stacking; a separate Random Forest on RGB/SIFT did not justify the added complexity.
Alternative classifiers	Random Forest, Extra Trees, CatBoost, Naive Bayes, NB-SVM, LR/MLP on embeddings, one-vs-rest and ordinal variants: tested as alternative prediction models but not selected.
Other kNN variants	Neighbours based on TF-IDF/numerical features and replacing SVM were inferior to adding embedding-based kNN alongside SVM.
Specialists for `below`	Binary classification, reweighting hard examples, predicting 0/1/2+ reactions: targeted low-engagement cases but produced no consistent overall macro-F1 gain.
SetFit / hard negatives	Contrastive fine-tuning of text embeddings, full text, pairs based on OOF errors and borderline examples: the gain did not justify the complexity.
Other encoders, OCR, supervised fine-tuning	English MiniLM, E5, MPNet, recognizing text in images and fine-tuning: the effect was small, absent or unconfirmed in further validation.
Network features	Link structure and neighbours based on organizations, intended to add social context: worse performance or minimal improvement.
Visual relationships and a conditional branch	Text/image similarity, visual history/novelty, a separate branch for posts with images: the added relationships and conditional processing did not justify the complexity.

RankNet	Learning pairwise rankings of posts within each author: a CV gain of about 0.00024 was too small.
Ensemble simplifications	Duplicate context in the meta-model was removed; removing images or history retrieval was rejected because performance deteriorated. These retained blocks still contribute useful information.
5 inner folds	Increasing the number of stacking folds: `evaluate` macro-F1 0.6618 → 0.6610; restored 3 folds. This is separate from the outer CV used to compare variants.

One issue affecting the metrics is authors with low popularity, for whom the reaction counts corresponding to different classes differ only slightly
(for example, when an author's typical reaction count is 2). Small absolute changes can then represent large relative changes, making the class boundary harder to predict from content. To address this issue, I tried various approaches (listed above),
but none produced a substantial improvement. SetFit is the most promising direction for further work (as well as, potentially, other algorithms for constructing hard negatives), although the tested versions did not justify inclusion in the final model.
The issue remains unresolved and requires further experimentation.

Final Metrics

The standard `evaluate.py`, the same test set: **17 623 posts, 824 authors**; train: 71 395 posts, 3 295 authors. The baseline uses post/author features without reading text or images. QWK accounts for class order, penalizing confusion between `below` and `above` more heavily.

Metric	Baseline	Final Model
macro-F1	**0.5410**	**0.6618**
QWK	0.4576	0.6037
Accuracy	0.5465	0.6991
F1 `below`	0.4859	0.5507
F1 `typical`	0.5689	0.7544
F1 `above`	0.5683	0.6802

The macro-F1 improvement over the baseline is **0.1208**. All three classes improve, although `below` remains the weakest class. Mean macro-F1 across **5 outer author-based CV folds - 0.6709**. Outer folds assess performance on held-out authors during development; the 3 inner folds generate stacking inputs. CV and the final test score refer to different evaluation splits.